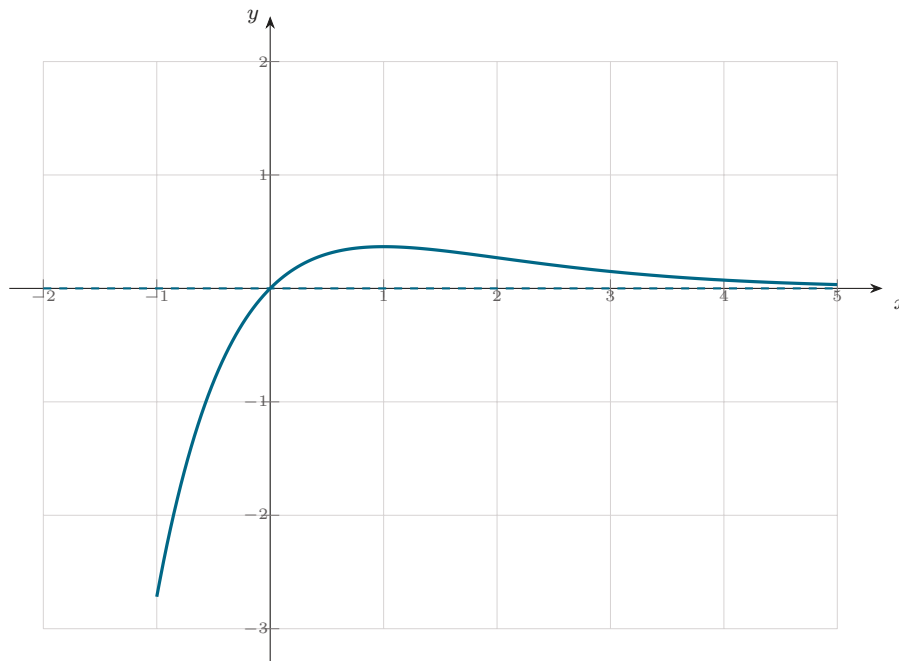


Corrigé — Exercice 18

A. 1) $\lim_{-\infty} g = -\infty$; $\lim_{+\infty} g = 0$ (croissance comparée, asymptote $y = 0$). **2)** $g'(x) = e^{-x} - xe^{-x} = (1-x)e^{-x}$; maximum $g(1) = \frac{1}{e}$. **3)** $G'(x) = -e^{-x} + (x+1)e^{-x} = xe^{-x} = g$. **B. 4)** $J_0 = \int_0^1 e^{-x} dx = 1 - \frac{1}{e}$; $J_1 = \int_0^1 xe^{-x} dx = [G]_0^1 = -\frac{2}{e} + 1 = 1 - \frac{2}{e}$. **5)** IPP ($u = x^{n+1}$, $dv = e^{-x} dx$, $v = -e^{-x}$): $J_{n+1} = [-x^{n+1}e^{-x}]_0^1 + (n+1)J_n = (n+1)J_n - \frac{1}{e}$. **6)** $J_{n+1} - J_n = \int_0^1 x^n(x-1)e^{-x} dx \leq 0$; intégrande ≥ 0 . **7)** $e^{-x} \leq 1$ sur $[0, 1]$ donne $J_n \leq \frac{1}{n+1}$, d'où $\lim J_n = 0$; et $(n+1)J_n = J_{n+1} + \frac{1}{e} \rightarrow \frac{1}{e}$.



$\mathcal{C}_g : y = xe^{-x}$ et l'asymptote $y = 0$ (Exercice 18).